

PRACTICAL WEEK10

TASK1: Investigate inode structures using commands such as `ls -li`, `stat`, and `find`. Create hard links and symbolic links and explain their impact on inode allocation.

```
laasya@laasya: ~/inode_lab
laasya@laasya:~$ mkdir inode_lab
laasya@laasya:~$ cd inode_lab
laasya@laasya:~/inode_lab$ echo "This is an inode practical file." > original.txt
laasya@laasya:~/inode_lab$ cat original.txt
This is an inode practical file.
laasya@laasya:~/inode_lab$ ls -li original.txt
57263 original.txt
laasya@laasya:~/inode_lab$ stat original.txt
  File: original.txt
  Size: 33          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 57263        Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/   laasya)   Gid: ( 1000/   laasya)
Access: 2026-09-27 11:01:35.229802669 +0000
Modify: 2026-09-27 11:01:29.617836307 +0000
Change: 2026-09-27 11:01:29.617836307 +0000
 Birth: 2026-09-27 11:01:29.617836307 +0000
laasya@laasya:~/inode_lab$ ln original.txt hardlink.txt
laasya@laasya:~/inode_lab$ ls -li original.txt hardlink.txt
57263 hardlink.txt 57263 original.txt
laasya@laasya:~/inode_lab$ stat original.txt
  File: original.txt
  Size: 33          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 57263        Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/   laasya)   Gid: ( 1000/   laasya)
Access: 2026-09-27 11:01:35.229802669 +0000
Modify: 2026-09-27 11:01:29.617836307 +0000
Change: 2026-09-27 11:01:56.681610400 +0000
 Birth: 2026-09-27 11:01:29.617836307 +0000
laasya@laasya:~/inode_lab$ ln -s original.txt symlink.txt
laasya@laasya:~/inode_lab$ ls -l
total 8
-rw-r--r-- 2 laasya laasya 33 Sep 27 11:01 hardlink.txt
-rw-r--r-- 2 laasya laasya 33 Sep 27 11:01 original.txt
lrwxrwxrwx 1 laasya laasya 12 Sep 27 11:02 symlink.txt -> original.txt
laasya@laasya:~/inode_lab$ ls -li original.txt hardlink.txt symlink.txt
57263 hardlink.txt 57263 original.txt 57265 symlink.txt
laasya@laasya:~/inode_lab$ stat symlink.txt
  File: symlink.txt -> original.txt
  Size: 12          Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48      Inode: 57265        Links: 1
```

```
laasya@laasya: ~/inode_lab
Change: 2026-09-27 11:01:56.681610400 +0000
Birth: 2026-09-27 11:01:29.617836307 +0000
laasya@laasya:~/inode_lab$ ln -s original.txt symlink.txt
laasya@laasya:~/inode_lab$ ls -l
total 8
-rw-r--r-- 2 laasya laasya 33 Sep 27 11:01 hardlink.txt
-rw-r--r-- 2 laasya laasya 33 Sep 27 11:01 original.txt
lrwxrwxrwx 1 laasya laasya 12 Sep 27 11:02 symlink.txt -> original.txt
laasya@laasya:~/inode_lab$ ls -i original.txt hardlink.txt symlink.txt
57263 hardlink.txt 57263 original.txt 57265 symlink.txt
laasya@laasya:~/inode_lab$ stat symlink.txt
  File: symlink.txt -> original.txt
  Size: 12          Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48      Inode: 57265          Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/  laasya)   Gid: ( 1000/  laasya)
Access: 2026-09-27 11:02:23.449349038 +0000
Modify: 2026-09-27 11:02:18.293386659 +0000
Change: 2026-09-27 11:02:18.293386659 +0000
Birth: 2026-09-27 11:02:18.293386659 +0000
laasya@laasya:~/inode_lab$ ls -l symlink.txt
lrwxrwxrwx 1 laasya laasya 12 Sep 27 11:02 symlink.txt -> original.txt
laasya@laasya:~/inode_lab$ find . -type f
./hardlink.txt
./original.txt
laasya@laasya:~/inode_lab$ echo "New line added." >> original.txt
laasya@laasya:~/inode_lab$ cat hardlink.txt
This is an inode practical file.
New line added.
laasya@laasya:~/inode_lab$ cat symlink.txt
This is an inode practical file.
New line added.
laasya@laasya:~/inode_lab$ rm original.txt
laasya@laasya:~/inode_lab$ ls
hardlink.txt  symlink.txt
laasya@laasya:~/inode_lab$ cat hardlink.txt
This is an inode practical file.
New line added.
laasya@laasya:~/inode_lab$ ls -l symlink.txt
lrwxrwxrwx 1 laasya laasya 12 Sep 27 11:02 symlink.txt -> original.txt
laasya@laasya:~/inode_lab$ ls -i
57263 hardlink.txt 57265 symlink.txt
```

TASK2: Design a program using mmap() to perform file reading and writing. Compare memory-mapped I/O with traditional read/write operations in terms of efficiency and implementation complexity.

laasya@laasya: ~/mmap_lab

mmap_file.c *

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main()
{
    int fd;
    struct stat file_stat;
    char *mapped;

    /* Open the file */
    fd = open("input.txt", O_RDWR);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    /* Get file size */
    if (fstat(fd, &file_stat) == -1)
    {
        perror("fstat");
        close(fd);
        return 1;
    }

    size_t file_size = file_stat.st_size;

    printf("File size: %ld bytes\n", file_size);

    /* Map the file into memory */
    mapped = mmap(NULL,
```

```

        return 1;
    }

    printf("\nOriginal file contents:\n");
    write(STDOUT_FILENO, mapped, file_size);

    /* Modify the mapped memory */
    const char *message = "Modified using mmap.";

    if (strlen(message) <= file_size)
    {
        memcpy(mapped, message, strlen(message));
    }
    else
    {
        printf("\nNew message is larger than the file.\n");
    }

    /* Make changes visible */
    if (msync(mapped, file_size, MS_SYNC) == -1)
    {
        perror("msync");
    }

    printf("\n\nFile modified using mmap().\n");

    /* Unmap memory */
    if (munmap(mapped, file_size) == -1)
    {
        perror("munmap");
    }

    close(fd);

    return 0;
}

```

```
laasya@laasya: ~/mmap_lab
laasya@laasya:~$ mkdir mmap_lab
laasya@laasya:~$ cd mmap_lab
laasya@laasya:~/mmap_lab$ echo "This file will be read and modified using mm
ap." > input.txt
laasya@laasya:~/mmap_lab$ v
v: command not found
laasya@laasya:~/mmap_lab$ cat input.txt
This file will be read and modified using mmap.
laasya@laasya:~/mmap_lab$ nano mmap_file.c
laasya@laasya:~/mmap_lab$ gcc mmap_file.c -o mmap_file
laasya@laasya:~/mmap_lab$ ./mmap_file
File size: 48 bytes

Original file contents:
This file will be read and modified using mmap.

File modified using mmap().
laasya@laasya:~/mmap_lab$ cat input.txt
Modified using mmap.ad and modified using mmap.
laasya@laasya:~/mmap_lab$ ls -l input.txt
-rw-r--r-- 1 laasya laasya 48 Sep 27 11:07 input.txt
laasya@laasya:~/mmap_lab$ stat input.txt
  File: input.txt
  Size: 48          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 57108         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/  laasya)   Gid: ( 1000/  laasya)
Access: 2026-09-27 11:07:47.198506968 +0000
Modify: 2026-09-27 11:07:33.666620937 +0000
Change: 2026-09-27 11:07:33.666620937 +0000
Birth: 2026-09-27 11:05:54.291477555 +0000
laasya@laasya:~/mmap_lab$ |
```